

Rahul Kalliadan

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Professional Summary

Senior Mechanical Engineer (Robotics) with 7+ years designing and launching Industrial AMRs and commercial cleaning robots. Delivered 25% unit cost reduction and 33% faster assembly via DFM/DFA, GD&T (ASME Y14.5) and supplier optimisation. Deployed 1.8k robots in service with IEC 60529 IPX3 validation and fleet reliability improvements.

Core Skills

- **Mechanical Design:** DFM/DFA, GD&T (ASME Y14.5), tolerance stacks, value engineering, design reviews
- **Manufacturing:** CNC, injection molding, die casting, vacuum casting, sheet metal, jigs/fixtures, Value Analysis & Value Engineering (VAVE)
- **Robotics/Systems:** AMR architecture, suspensions, lifting/tug modules, cleaning mechanisms, docking stations, IP sealing, sensor configuration
- **Analysis/Software:** SolidWorks PDM/3DEXperience, Ansys (FEA/CFD), Matlab, ROS, SolidWorks Composer
- **Quality & Compliance:** DFMEA, ECNs, IQC/OQC, IEC 60529 (IPX3), ISO 12100, ISO 45001
- **Project & Ops:** Agile workflows (Confluence, Notion), BOM & cost engineering, cross-team collaboration, ECN, supplier coordination

Experience

Senior Mechanical Engineer, Lionsbot International Pte Ltd – Singapore **Apr 2022 – Jul 2025**

- Owned mechanical design and system architecture for **R3 Scrub Pro** from concept to production; shipped 1.8k+ units across multiple markets, achieving IPX3 (IEC 60529) ingress compliance.
- Delivered 25% part cost reduction via design consolidation and supplier process changes; reduced assembly time 33% through DFA and fixture redesign.
- Designed, engineered, and validated handlebar mechanism, cleaning mechanism, vacuum chamber and safety bumper subsystems.
- Managed cross-functional timelines, supplier handovers, and production SOPs, enabling smooth transition from prototype builds to mass manufacturing; coordinated design → validation → ramp with QA and production teams.
- Developed suspension mechanism (12deg slope capability) and chassis design using first-principles engineering (traction, spring, and stability calculations); completed full CAD integration, sensor layout, and water-line system, enabling prototype readiness.

Senior Mechanical Engineer, Botsync Technologies – Bangalore, India **Jul 2021 – Mar 2022**

Mechanical Engineer, Botsync Technologies – Bangalore, India **Mar 2020 – Jun 2021**

- Sole owner of all mechanical development for **MAG500 and MAG1000** AMRs (500 kg & 1000 kg payloads), from architecture and subsystem design to validation and production release.
- Designed and delivered chassis, suspension, pallet lifter, tug modules, and docking stations, enabling payload handling and logistics automation across multiple client use-cases.
- Led CAD modelling, tolerance stacks, and FEA validation, ensuring structural robustness and reliability for warehouse and factory operations.
- Set up and standardised PDM and release workflows, reducing design cycle time and establishing traceability across BOMs and ECNs.
- Owned supplier engagement and prototype builds, translating designs into manufacturable assemblies.
- Represented mechanical function in client pilots and DARs (e.g., TVS Motor, Ford Manufacturing Unit); resolved all CAPAs within 2 weeks, enabling POC approval and customer adoption.

Solution Design Engineer, Addverb Technologies – Noida, India **Jan 2019 – Feb 2020**

- Owned mechanical design for AMR pilot projects with FMCG clients (e.g., HUL Ice Cream, Lifebuoy, Cofresh); delivered functional prototypes and BOM packages that achieved 100% first-pass site acceptance.

- Designed conveyor systems (belt conveyors, roller conveyors) for automated material handling; optimised layouts for throughput and space utilisation.
- Commissioned conveyor unit at HUL site, overseeing installation and running system for 3 weeks without operational issues; ensured uptime and client satisfaction during handover.
- Supported other project teams with manufacturing drawings, applying GD&T and drawing standards to ensure quality, manufacturability, and vendor clarity.
- Coordinated vendor development and prototyping, translating designs into manufacturable units within cost and schedule targets.
- Collaborated with cross-functional teams (controls, project managers) to deliver integrated automation solutions aligned with client operations.

Mechanical Engineer, Airlitix Inc – California (Remote Work)

Nov 2017 – Dec 2018

- Performed CFD simulations in Ansys to evaluate duct geometries for greenhouse drone applications, selecting designs that maximised thrust efficiency while minimising losses.
- Contributed to the initial prototype design of greenhouse drones, supporting CAD modelling, integration of ducted fan systems, and early proof-of-concept builds.
- Collaborated with the computer vision team, providing mechanical input for sensor placement and supporting algorithm validation.
- Delivered design recommendations that informed subsequent prototype iterations, enabling a more efficient drone architecture tailored for greenhouse environments.

Mechanical Intern, Ernst & Young – Trivandrum, India

Aug 2017 – Oct 2017

- Contributed to early-stage design of an underwater robot for bridge inspection and water-quality monitoring.
- Explored advanced technologies, including Microsoft HoloLens for visualisation and ROS-based computer vision pipelines.

Education

Amrita School of Engineering, Master of Technology - Robotics and Automation

Jul 2015 - Jun 2017

Visvesvaraya Technological University, Bachelor of Engineering - Mechanical Engineering

Jul 2011 - Jun 2015

Publications

- Design and validation of low-cost wearable assistive device for carrying back loads, IConAMMA-2017
- Low cost mechanical assistive aid for workers carrying head loads, TIAR 2017 (IEEE Xplore)